

Human Health and Disease

1. Saliva in mouth and mucus coating of epithelium

lining the urogenital tract form

- (1) Physical barriers of innate immunity
- (2) Physiological barriers of innate immunity
- (3) Physical and physiological barriers of innate immunity respectively
- (4) Physiological and physical barriers of innate immunity respectively

2. Choose the correct answer from the options given below

Statement-I : Plasmodium, a tiny protozoan causes malaria.

Statement-II : Chill and high fever recurring every three to four days in malaria.

- (1) Both Statement-I and Statement-II are incorrect.
- (2) Statement-I is correct but statement-II is incorrect.
- (3) Statement-I is incorrect but Statement-II is correct.
- (4) Both Statement-I and Statement-II are correct.

3. Which of the following is not correct about pneumonia

- (a) It is caused by streptococcus pneumoniae
- (b) Infects the alveoli of the lungs
- (c) Symptoms of pneumonia include fever, chills, cough and headache
- (d) In all cases the lips and finger nails turn blue to grayish in colour

Select the correct answer

- (1) a, b, d
- (2) b & d
- (3) d only
- (4) a, b, c, d

4. Match the column :

Column-I

Column-II

(a)Morphine

(i) Hallucinogen

(b)LSD

(ii) Stimulant

(c)Marijuana

(iii) Sedative & analgesic

(d)Coke

(iv) Cannabis sativa

(1) a-(iii), b-(i), c-(iv), d-(ii)

(2) a-(i), b-(ii), c-(iv), d-(iii)

(3) a-(ii), b-(i), c-(iii), d-(iv)

(4) a-(iii), b-(i), c-(ii), d-(i)

5. Health, for a long time, was considered as a state of body and mind where there was a balance of certain

- (1) Antigens
- (2) Pathogens
- (3) Humors
- (4) Allergens

6. Complete the analogy by selecting the correct option w.r.t. diseases and the symptoms associated with them in severe cases.

Intestinal perforation and death : Typhoid :: Finger nails and lips turn gray to bluish in colour : ____.

- (1) Ringworm
- (2) Pneumonia
- (3) Ascariasis
- (4) Common cold

7. The yellowish fluid secreted by mammary glands of mother during the initial days of lactation has abundant

- (1) IgM
- (2) IgA
- (3) IgG
- (4) IgE

8. Observe the figure given below and select the correct option w.r.t. drug obtained from it.



- (1) Generally taken by injection
(2) Known for their effect on cardio-vascular system
(3) Effective sedative and pain killer
(4) Receptors present only in GIT

9. Select the correct set of correct statements from the options given below.

- (A) Haemophilus influenzae is responsible for pneumonia disease.
(B) Common cold caused by Rhino virus.
(C) Haemocoin responsible for the chill & high fever in malaria.
(D) Virus infected cells secrete protein called interferons.

- (1) Only A & B are correct
(2) Only B & C are correct
(3) Only C & D are correct
(4) Only A, B & D are correct

10. Malignant neoplasm have all of the following characteristics, except

- (1) Uncontrolled mitosis
(2) Contact inhibition
(3) Compete with normal cells for nutrients
(4) Invasion of normal tissues

11. Though all persons are vulnerable to STDs, their incidences are reported to be very high among persons in the age group of :

- (1) 15 - 24 years
(2) 25 - 45 years
(3) 1 - 5 years
(4) 45 - 75 years

12. Disease are broadly grouped into infectious and non infectious diseases. In the list given below, identify the infectious diseases :

- i. Cancer ii. Influenza
iii. Allergy iv. Small pox

- (1) i and ii
(2) ii and iii
(3) iii and iv
(4) ii and iv

13. Select the option representing only the correct examples of passive immunity.

- a. Immunoglobulin transmitted through mother's milk
b. Immunoglobulin injected in a patient
c. Attenuated pathogens introduced in the body
d. Administration of anti-venom

- (1) a, b, c and d
(2) a, b and d
(3) a and d only
(4) b and d only

14. In an antibody molecule, the antigen binding site is formed at

- (1) N-terminal of light chain only
(2) N-terminal of heavy chain only
(3) N-terminal of both heavy and light chains
(4) C-terminal of both heavy and light chains

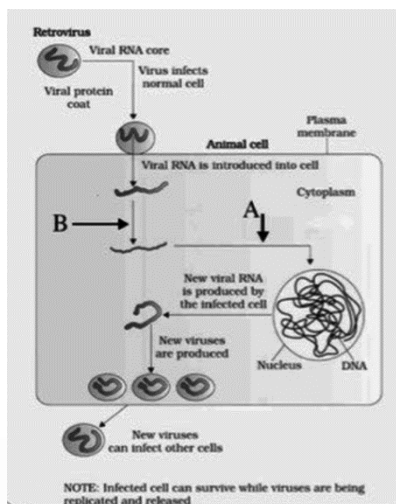
15. Given below are two statements one is labelled as Assertion (A) and other labelled as Reason (R).

Assertion (A) : HIV is a retrovirus.

Reason (R) : RNA genome of the virus replicates to form viral DNA with the help of enzyme reverse transcriptase.

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
(2) Both (A) and (R) are correct and (R) is the correct explanation of (A).
(3) (A) is correct but (R) is not correct.
(4) (A) is not correct but (R) is correct.

16.



Identify the events indicated by A and B in given figure.

- (1) A → Viral RNA is produced by reverse transcriptase
- (2) B → Viral RNA incorporates into host genome
- (3) A → Viral DNA is produced by reverse transcriptase
- (4) B → Host DNA incorporates into viral genome.

17. Transmission of HIV :

- (a) Transfusion of contaminated blood.
- (b) Sexual contact with infected person
- (c) From infected mother to her baby through placenta.
- (d) Sharing infected needles
- (e) Sharing food with infected person

Which statements given above is/are not correct ?

- (1) a, b, e
- (2) e only
- (3) a, c, d, e
- (4) a, b, c, d

18. A person who was bitten by a venomous snake was taken to a hospital for immediate treatment where he was injected with a solution which contained

- (1) Attenuated pathogens
- (2) Pre-formed antigens
- (3) Active antigens
- (4) Pre-formed antibodies

19. Match the Column-I with Column-II

(a)	Amoebiasis	(i)	Blockage of intestinal passage
(b)	Malaria	(ii)	Stools with excess mucous & blood clots
(c)	Ascariasis	(iii)	Release of haemozoin
(d)	Filariasis	(iv)	Gross deformities in genital organs

- (1) a(i), b(ii), c(iii), d(iv)
- (2) a(iv), b(iii), c(ii), d(i)
- (3) a(iii), b(i), c(ii), d(iv)
- (4) a(ii), b(iii), c(i), d(iv)

20. Virus-infected cells secrete proteins which protect non infected cells from further viral infection. These proteins constitute/form which type of barrier of innate immunity?

- (1) Physical
- (2) Cellular
- (3) Cytokine
- (4) Physiological

21. Where immature lymphocytes differentiate into antigen sensitive lymphocyte :

- (1) M.A.L.T
- (2) Primary lymphoid organs
- (3) Secondary lymphoid organs
- (4) Spleen

22. Choose correct answer from options given below

Statement-I : *Salmonella typhi* causes typhoid fever.

Statement-II : Typhoid fever could be conformed by Widal test.

- (1) Both Statement-I and Statement-II are incorrect.
- (2) Statement-I is correct but statement-II is incorrect.
- (3) Statement-I is incorrect but Statement-II is correct.
- (4) Both Statement-I and Statement-II are correct.

23. Assertion:- Emphysema is caused by cigarette smoking.

Reason:- Cigarette smoking increases the gaseous exchange area.

- (1) If both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.
- (3) If Assertion is True but the Reason is False.
- (4) If both Assertion & Reason are False.

24. Below are some statements related to life cycle of Plasmodium. Which of the following is correct?

- (1) Gametocytic stage infects human body
- (2) It multiplies by sexual method of reproduction in human liver
- (3) human RBC are ruptured and releases a toxin hemozoin
- (4) development of gametocyte possible only in female culex mosquito

25. The use of vaccines and immunization programmes have enabled us to completely eradicate a deadly disease called

- (1) Diphtheria
- (2) Cancer
- (3) Pneumonia
- (4) Small pox

26. Choose correct statement among the following.

- (1) Ringworm is a protozoal disease caused by Salmonella.
- (2) Malignant malaria is caused by P. vivax.
- (3) Crack is obtained from the poppy plant.
- (4) Microsporum is the name of a fungal genus.

27. The side-effects of the misuse of anabolic steroids in human females do not include

- (1) Masculinisation
- (2) Enlargement of clitoris
- (3) Breast enlargement
- (4) Abnormal menstrual cycles

28. Select the incorrect option w.r.t. techniques used for the diagnosis of cancer.

- (1) Biopsy – A piece of suspected tissue cut into thin section is stained and examined under microscope
- (2) Radiography – Use of X-rays is involved
- (3) MRI – Involves use of strong magnetic fields and ionising radiations to accurately detect physiological changes in the living tissue
- (4) Computed tomography – Involves use of X-rays to generate a three-dimensional image of the internals of an object

29. Given below are two statements one is labelled as Assertion (A) and the other labelled as Reason (R).

Assertion (A): Addiction to smack is rampant in adolescents present in colleges and schools to increase energy and producing sense of euphoria, prior to examination.

Reason (R): Diacetylmorphine has a potent stimulating action on central nervous system.

In the light of above statements, select the correct option.

- (1) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (2) (A) is true but (R) is false
- (3) Both (A) and (R) are false
- (4) Both (A) and (R) are true but (R) is not the correct explanation of (A)

30. Smack is a white, odourless, bitter crystalline drug which is obtained by the _____ of morphine. Select the option to fill the blank correctly.

- (1) Methylation
- (2) Polymerisation
- (3) Acetylation
- (4) Nitration

31. Choose the correct option about given picture



A

B

- | | |
|------------------|------------|
| (1) Hallucinogen | Depressant |
| (2) Morphine | Datura |
| (3) Heroin | Bhang |
| (4) Datura | Morphine |

32. Read the following statements

- (A) The macrophage continue to produce virus and in this way like HIV factory.
- (B) α interferon which activates immune system and help in destroying the tumor
- (C) Morphine extracted from the latex of poppy plant.
- (D) Cocaine is obtained from Coca plant

Erythroxylum Coca.

Select the correct set of correct statements from the options given below.

- (1) A & B are correct
- (2) B & C are correct
- (3) C & D are correct
- (4) A, B, C & D are correct

33. Opioids are the drugs which bind to specific

opioid receptors present in our CNS and GIT.

Heroin, commonly called as smack is chemically....., which is white, odourless, bitter crystalline compound, obtained by.....of morphine ?

- (1) Diacetylmorphine, methylation
- (2) Diacetylmorphin, acetylation
- (3) Benzodiazepines, animation
- (4) Amphetamines, acetylation

34. Choose the correct answer from the options given below

Statement-I : Hippocrates and Indian Ayurveda system of medicine thought that persons with "black bile" belonged to hot personality and would have fever.

Statement-II : William Harvey disproved good humor hypothesis of health.

- (1) Both Statement-I and Statement-II are incorrect.
- (2) Statement-I is correct but statement-II is incorrect.
- (3) Statement-I is incorrect but Statement-II is correct.
- (4) Both Statement-I and Statement-II are correct.

35. In which disease does mosquito transmitted pathogen cause chronic inflammation of lymphatic vessels ?

- (1) Elephantiasis
- (2) Rigworm
- (3) Ascariasis
- (4) Amoebiasis

36. When the same antigen or pathogen enters the human body for the second time, the resultant secondary immune response is stronger than primary immune response due to the presence of

- (1) Neutrophils only
- (2) T-helper cells only
- (3) Memory cells
- (4) Macrophages

37. If the immune system of higher vertebrates such as humans loses the property of distinguishing self-cells from pathogens due to some unknown reasons and thereby attacks self-cells, then it is called

- (1) Allergic response
- (2) Graft rejection
- (3) Acquired immunity
- (4) Autoimmunity

38. Read the given statements A and B carefully and select the most appropriate option.

Statement A: Gametocyte of Plasmodium is the infective stage for human host.

Statement B: Sporozoite of Plasmodium is the infective stage for the vector.

- (1) Both statements A and B are correct
- (2) Only statement A is incorrect
- (3) Only statement B is incorrect
- (4) Both statements A and B are incorrect

39. Which of the following is the site of differentiation of immature lymphocytes into antigen-sensitive lymphocytes?

- (1) Bone marrow
- (2) Spleen
- (3) Lymph node
- (4) Tonsil

40. Identify the drugs which are misused by certain sportspersons to enhance their performance and choose the correct option.

- I. Narcotic analgesics II. Diuretics
- III. Anabolic steroids IV. Amphetamines
- (1) III and IV only
- (2) I and II only
- (3) I, III and IV only
- (4) I, II, III and IV

41. Identify the correct pair representing the causative agent of typhoid fever and the confirmatory test for typhoid:

- (1) Salmonella typhi/ Widal test
- (2) Plasmodium vivax / UTI test
- (3) Streptococcus pneumoniae/ Widal test
- (4) Salmonella typhi / Anthrone test

42. Which is an auto immune disease?

- (1) Myasthenia gravis
- (2) SCID
- (3) Asthma
- (4) Alzheimer disease

43. Choose the correct answer from the options given below

Statement-I : Virus infected cells secrete protein called interleukins.

Statement-II : Interleukins protein protect non infected cell from further viral infection

- (1) Both Statement-I and Statement-II are incorrect.
- (2) Statement-I is correct but statement-II is incorrect.
- (3) Statement-I is incorrect but Statement-II is correct.
- (4) Both Statement-I and Statement-II are correct.

44. Select the correct set of correct statements from the options given below.

- (A) Morphine is a very effective sedative and pain killer.
- (B) Cannabinoid receptor is mainly found in brain.
- (C) Opioid receptors are present in CNS and GIT.
- (D) Barbiturates are used as medicine to help patient cope with insomnia.

- (1) A & B are correct
- (2) B & C are correct
- (3) C & D are correct
- (4) A, B, C & D are correct

45. Match the following :

(A)	Fever	(i)	Cellular immunity
(B)	Monocyte	(ii)	humoral immunity
(C)	Killer T-cells	(iii)	Physiological barrier
(D)	Plasma cells	(iv)	Cellular barrier

Choose the correct option

- (1) A-(iv), B-(i), C-(iv), D-(ii)
- (2) A-(iii), B-(iv), C-(i), D-(ii)
- (3) A-(ii), B-(iv), C-(i), D-(ii)
- (4) A-(ii), B-(i), C-(iv), D-(iii)

46. The cancer patients are given specific substances which activate their immune system and help in destroying the tumor. These substances are

- a. Biological response modifiers**
- b. α -interferons**
- c. Hallucinogens**

Select the correct option

- (1) a and c
- (2) b and c
- (3) a and b
- (4) a, b and c

47. Given below are two statements one is labelled as Assertion (A) and the other labelled as Reason (R).

Assertion (A) : In severe cases of pneumonia, the lips and finger nails may turn gray to bluish in colour.

Reason (R) : In pneumonia the alveoli get filled with fluid leading to severe problems in respiration.

- (1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).
- (2) Both (A) and (R) are correct and (R) is the correct explanation of (A).
- (3) (A) is correct but (R) is not correct.
- (4) (A) is not correct but (R) is correct.

48. Consider the following statements and select the option which correctly represents them as true(T) or false(F).

- a. The use of histamine and adrenaline quickly reduces the symptoms of allergy.**
- b. Hashish is a depressant, obtained from poppy plant.**

c. Smoking reduces the concentration of haembound oxygen in blood

d. The tolerance level of receptors present in our body decreases with the repeated use of drugs.

- (1) a(F), b(T), c(T), d(T)
- (2) a(F), b(F), c(T), d(T)
- (3) a(F), b(F), c(F), d(T)
- (4) a(F), b(F), c(T), d(F)

49. Select the correct set of correct statements from the options given below.

(A) Primary lymphoid organs are bone marrow and thymus.

(B) Hepatitis B vaccine is produced from yeast.

(C) MALT constitutes about 50% of the Lymphoid tissue in human body.

(D) AIDS is caused by HIV.

- (1) A & B are correct
- (2) B & C are correct
- (3) C & D are correct
- (4) A, B, C & D are correct

50. Read the following statements.

a. It was first reported in 1983.

b. It is caused by an enveloped rhinovirus.

c. Transmission of pathogen generally occurs through inhaling aerosols.

d. Treatment with anti-retroviral drugs is only partially effective.

How many of the statements mentioned above is/are true w.r.t. AIDS?

- (1) One
- (2) Two
- (3) Three
- (4) Four

Answer Key

1. (4)
2. (4)
3. (3)
4. (1)
5. (3)
6. (2)
7. (2)
8. (2)
9. (4)
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41. (1)
42. (1)
43. (1)
44. (4)
45. (2)



- 46. (3)
- 47. (2)
- 48. (4)
- 49. (4)
- 50. (1)

